Projectiles

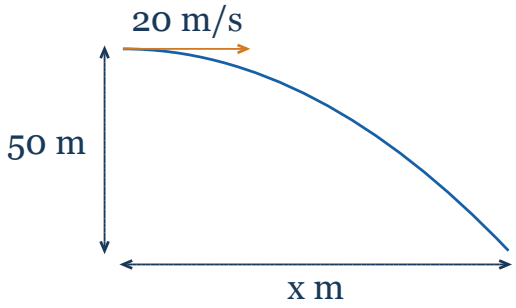
Solutions

Overview

In Mechanics Year 1, vertical projection used the suvat equations with acceleration $g\,\mathrm{m\,s^{-2}}$ downwards. Here objects can also be projected sideways.

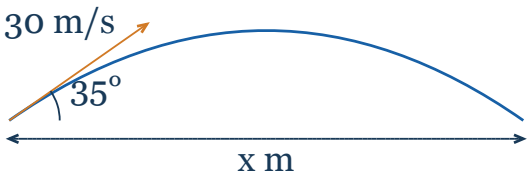
1: Horizontally projected

A particle is projected horizontally at $20\,\mathrm{m\,s^{-1}}$, 50 m above the ground. How far along the ground does it travel?



2: Projection at any angle

A cabbage is projected from ground level at $30\,\mathrm{m\,s^{-1}}$, at 35° to the horizontal. How far away does it hit the ground?



Acceleration in Each Direction

Solution 3

Consider vertical and horizontal motion separately.

Vertical acceleration downwards:

$g\,\mathrm{m\,s^{-2}}$. Use the suvat equations.

Horizontal acceleration:

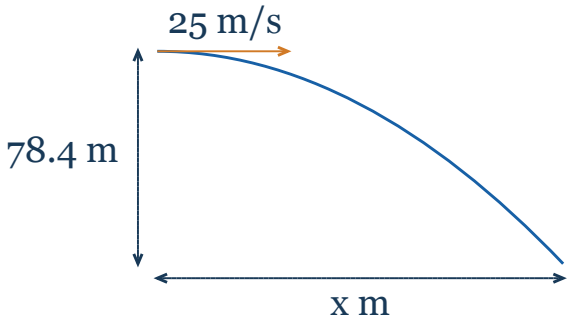
$0\,\mathrm{m\,s^{-2}}$. Velocity is constant, so speed = distance/time.

A particle is projected horizontally at $25\,\mathrm{m\,s^{-1}}$ from 78.4 m above a horizontal surface. Find:

a. The time to reach the surface.

b. The horizontal distance travelled.

c. The distance of the impact point from the original point.



a

Take downwards as positive:

$s = 78.4, \quad u = 0, \quad a = 9.8.$

$s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad 78.4 = 4.9t^2.$

$t = 4\,\mathrm{s}.$

b

Horizontally, use $s = vt$:

$x = 25 \times 4 = 100\,\mathrm{m}.$

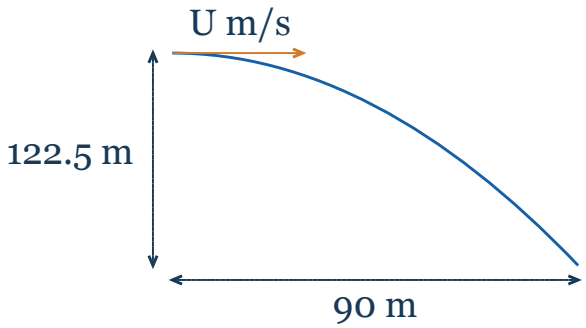
c

$\sqrt{78.4^2 + 100^2} = 130\,\mathrm{m} \quad (2\text{ s.f.}).$

Further Example

Solution 4

A particle is projected horizontally at $U\,\mathrm{m\,s^{-1}}$ from 122.5 m above a horizontal plane. It lands 90 m horizontally from its starting point. Find its initial speed.



Working

Take downwards as positive:

$s = 122.5, \quad u = 0, \quad a = 9.8, \quad t$ unknown.

$s = ut + \frac{1}{2}at^2$

$122.5 = 4.9t^2 \quad \Rightarrow \quad t = 5\,\mathrm{s}.$

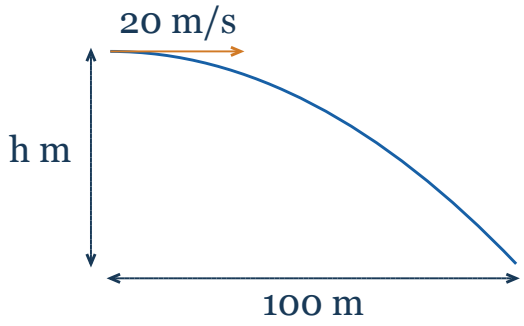
Horizontally:

$90 = U \times 5 \quad \Rightarrow \quad U = 18\,\mathrm{m\,s^{-1}}.$

Test Your Understanding

Solution 5

A particle is projected horizontally at $20\,\text{m s}^{-1}$ from h m above a horizontal plane. It lands 100 m horizontally from its starting point. Determine h.



Working

Horizontally:

$$t = \frac{100}{20} = 5\,\text{s}.$$

Taking downwards as positive:

$$s = h, \quad u = 0, \quad a = 9.8, \quad t = 5.$$

$$s = ut + \tfrac{1}{2}at^2$$

$$h = \tfrac{1}{2} \times 9.8 \times 5^2 = 122.5\,\text{m}.$$

Exercise 6A — Y2 Mechanics Pages 110–111

Solution 6

5. A particle moves horizontally at $20\,\text{m s}^{-1}$ along a flat smooth table, starting 2 m from its edge. The table is 1.2 m above the floor. Find the total time from projection until it lands on the floor.

Working

On the table, horizontally:

$$s = vt \quad \Rightarrow \quad 2 = 20t \quad \Rightarrow \quad t = 0.1\,\text{s}.$$

After leaving the table, taking downwards as positive:

$$u_y = 0, \quad a = 9.8, \quad s = 1.2.$$

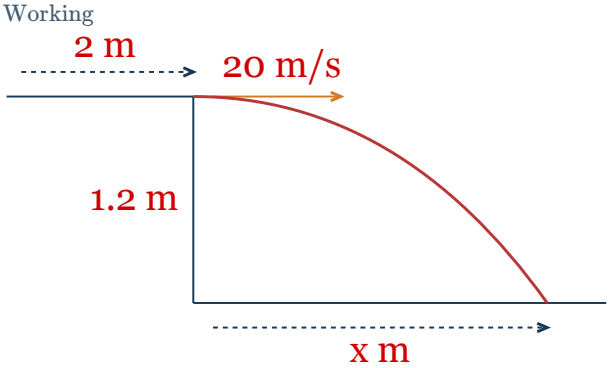
$$s = ut + \tfrac{1}{2}at^2 \quad \Rightarrow \quad 1.2 = 4.9t^2.$$

$$t^2 = \frac{1.2}{4.9}, \quad t = 0.49487\ldots\,\text{s}.$$

Reject the negative root. Total time:

$$0.1 + 0.49487\ldots = 0.59\,\text{s} \quad (2\text{ s.f.}).$$

Working



Exercise 6A — Y2 Mechanics Pages 110–111 Solution 7

6. A dart is thrown horizontally at $14\,\text{m s}^{-1}$ towards a vertical dartboard. Model the dart as a particle moving freely under gravity, in a vertical plane perpendicular to the board. It hits 9 cm below its launch level. Find the horizontal distance from the launch point to the board.

Working

Take downwards as positive:

$u_y = 0, \quad a = 9.8, \quad s = 0.09\,\text{m}.$

$s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad 0.09 = 4.9t^2.$

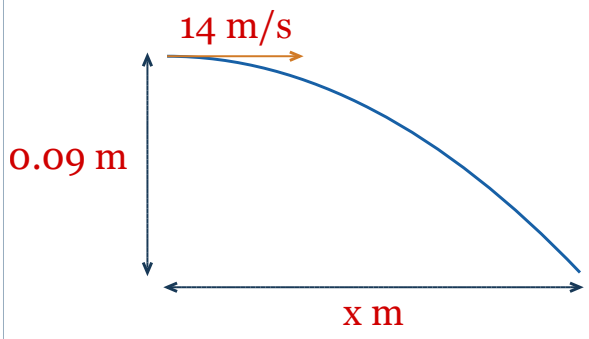
$t^2 = \dfrac{0.09}{4.9}, \quad t = 0.13552\ldots\ \text{s}.$

Reject the negative root. Horizontally, $u_x = 14$:

$s = vt \quad \Rightarrow \quad x = 14 \times 0.13552\ldots$

$x = 1.8973\ldots = 1.90\,\text{m} \quad (3\ \text{s.f.}).$

Working



Exercise 6A — Y2 Mechanics Pages 110–111 Solution 8

7. A particle of mass 2.5 kg is projected along a horizontal rough surface at $5\,\text{m s}^{-1}$. After 2 m, it leaves as a projectile and lands 1.2 m below. Total time from projection to landing is 1.0 s. Find:

a. The time in contact with the surface.

b. The coefficient of friction between particle and surface.

a

After leaving the surface, take downwards as positive:

$u_y = 0, \quad a = 9.8, \quad s = 1.2.$

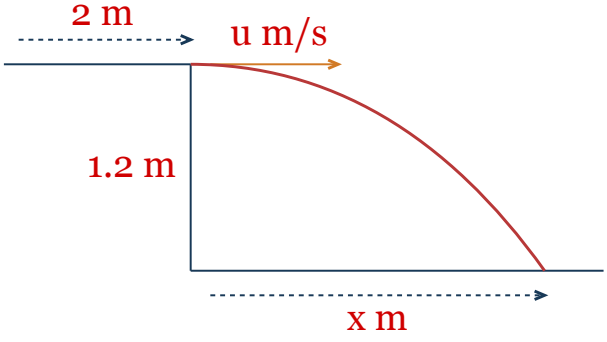
$s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad 1.2 = 4.9t^2.$

$t = \sqrt{1.2/4.9} = 0.49487\ldots\ \text{s}.$

Time on surface:

$1 - 0.49487\ldots = 0.50513\ldots\ \text{s}.$

a



b

On the surface, vertically $R = mg$. Horizontally:

$F = ma \quad \Rightarrow \quad -\mu R = ma.$

$-\mu mg = ma \quad \Rightarrow \quad a = -\mu g.$

With $s = 2$, $u = 5$, $t = 0.50513\ldots$:

$s = ut + \frac{1}{2}at^2$

$2 = 5(0.50513\ldots) + \frac{1}{2}a(0.50513\ldots)^2$

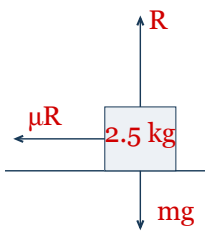
$0.12757\ldots a = 2 - 2.5256\ldots$

$a = -4.1201\ldots\ \text{m s}^{-2}.$

$-4.1201\ldots = -9.8\mu \quad \Rightarrow \quad \mu = 0.42042\ldots$

$\mu = 0.42 \quad (2\ \text{s.f.}).$

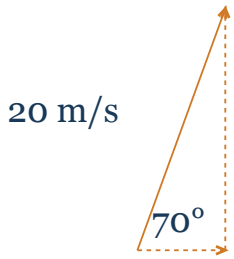
b



Components of Velocity

Solution 9

As with forces, split velocity into horizontal and vertical components.

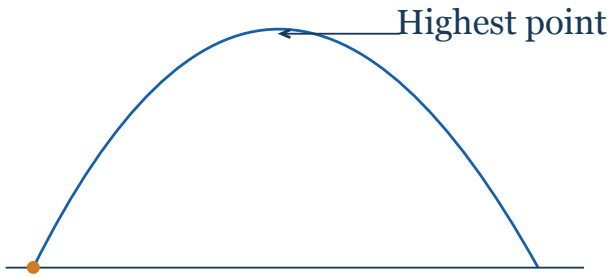


Horizontal component:

$20\cos 70^\circ \text{ m s}^{-1}$.

Vertical component:

$20\sin 70^\circ \text{ m s}^{-1}$.



At the highest point, the vertical velocity is:

0.

Speed is the magnitude of velocity. Initially:

$\mathbf{v} \approx \begin{pmatrix} 6.84 \\ 18.8 \end{pmatrix} \text{ m s}^{-1}$.

$$|\mathbf{v}| = \sqrt{(20\cos 70^\circ)^2 + (20\sin 70^\circ)^2}$$
$$\approx \sqrt{6.84^2 + 18.8^2} \approx 20 \text{ m s}^{-1}.$$

Simple Example

Solution 10

A particle P is projected from O on a horizontal plane at 28 m s^{-1}, at elevation 30°. It moves freely under gravity until it strikes the plane at A. Find:

a. Its greatest height above the plane.

b. Its time of flight.

c. The distance OA.

a

Take upwards as positive:

$u = 28\sin 30^\circ = 14, \quad v = 0, \quad a = -9.8.$

$v^2 = u^2 + 2as \quad \Rightarrow \quad 0 = 14^2 + 2(-9.8)s.$

$s = 10 \text{ m}.$

b

Back at its original level, vertical displacement is 0:

$s = 0, \quad u = 14, \quad a = -9.8.$

$s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad 0 = 14t - 4.9t^2 = t(14 - 4.9t).$

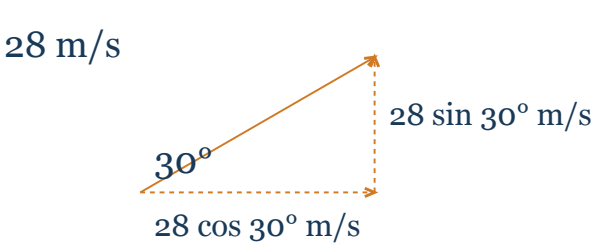
$t = 0 \quad \text{or} \quad t = \dfrac{14}{4.9} = 2.857\ldots \text{ s}.$

Time of flight is 2.9 s (2 s.f.).

c

$OA = 28\cos 30^\circ \times 2.857\ldots$

$= 69.282\ldots = 69 \text{ m} \quad (2 \text{ s.f.}).$



Projected from Above Ground

Solution 11

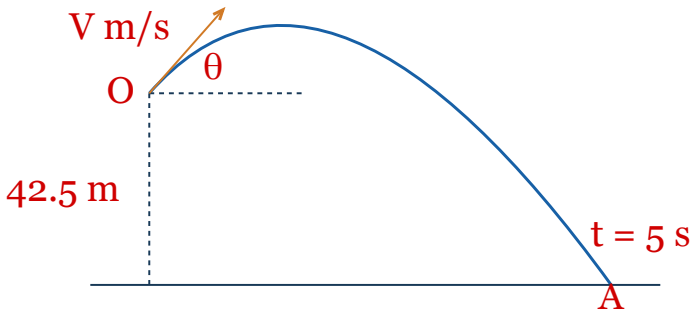
A particle is projected from O at $V\ \text{m s}^{-1}$, at elevation θ, where $\tan\theta = 4/3$. Point O is 42.5 m above a horizontal plane. The particle strikes the plane at A after 5 s.

a. Show that $V = 20\ \text{m s}^{-1}$.

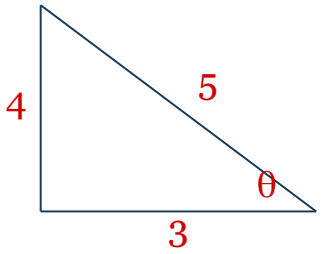
b. Find OA.

Draw a diagram before solving.

Diagram



a



a

$\sin\theta = \dfrac{4}{5}, \qquad \cos\theta = \dfrac{3}{5}.$

Take upwards as positive:

$s = -42.5, \quad u = \dfrac{4}{5}V, \quad a = -9.8, \quad t = 5.$

$s = ut + \frac{1}{2}at^2$

$-42.5 = \dfrac{4}{5}V \times 5 - 4.9 \times 25$

$V = 20\ \text{m s}^{-1}.$

b — horizontal

Let the horizontal distance be x:

$x = 20\cos\theta \times 5 = \dfrac{3}{5}(20) \times 5 = 60\ \text{m}.$

b — distance

By Pythagoras:

$OA = \sqrt{42.5^2 + 60^2} = 73.527\ldots$

$OA = 74\ \text{m} \quad (2\ \text{s.f.}).$

Time Above a Given Point

Solution 12

A particle is projected from O at $35\ \text{m s}^{-1}$, at elevation $30°$. It moves freely under gravity. Find the length of time for which it is 15 m or more above O.

Find the two times at height 15 m; subtract them.

Working

Take upwards as positive:

$s = 15, \quad u = 35\sin 30° = 17.5, \quad a = -9.8.$

$s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad 15 = 17.5t - 4.9t^2.$

$4.9t^2 - 17.5t + 15 = 0.$

Use the quadratic solver on your calculator:

$t = \dfrac{10}{7}, \quad \dfrac{15}{7}.$

Time at or above 15 m:

$\dfrac{15}{7} - \dfrac{10}{7} = \dfrac{5}{7} \approx 0.71\ \text{s} \quad (2\ \text{s.f.}).$

The supplied textbook permits $5/7$; the model uses approximate $g = 9.8$, so 0.71 s expresses the physical precision.

Test Your Understanding — Solution 13

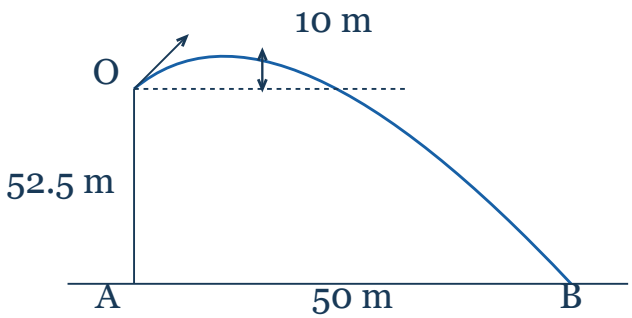
Edexcel M2 (old), May 2012, Q7

A small stone is projected from O, at the top of a vertical cliff OA, 52.5 m above the sea. It rises 10 m above O, then hits the sea at B, with $AB = 50$ m. Model it as a particle moving freely under gravity.

a. Show its initial vertical velocity component is $14\,\text{m s}^{-1}$. (3 marks)

b. Find its speed of projection. (9 marks)

c. Find the time when it moves parallel to OB. (5 marks)



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

At maximum height, $v_y = 0$:

$0 = u_y^2 - 2(9.8)(10) \quad \Rightarrow \quad u_y = 14\,\text{m s}^{-1}.$

b

Upwards positive:

$-52.5 = 14t - \frac{1}{2}gt^2$

$49t^2 - 140t - 525 = 0$

$(t - 5)(49t + 105) = 0 \quad \Rightarrow \quad t = 5\,\text{s}.$

Horizontally, $50 = 5u_H$, so $u_H = 10$.

$u = \sqrt{10^2 + 14^2} = \sqrt{296} = 17.2\,\text{m s}^{-1}.$

c

$\tan \angle OBA = \dfrac{52.5}{50} = 1.05.$

When parallel to OB, the downward vertical speed is $1.05 \times 10 = 10.5$.

$-10.5 = 14 - gt \quad \Rightarrow \quad t = 2.5\,\text{s}.$

Exercise 6B — Y2 Mechanics Page 113 — Solution 14

7. A particle is projected with initial velocity $\mathbf{U} = 3k\mathbf{i} + 2k\mathbf{j}\,\text{m s}^{-1}$.

a. Find the angle of projection.

b. Given its initial speed is $3\sqrt{13}\,\text{m s}^{-1}$, find k.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

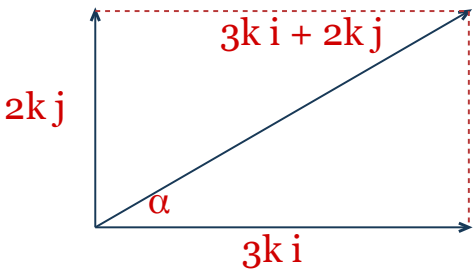
Let the angle of the velocity line be α:

$\tan\alpha = \dfrac{2k}{3k} = \dfrac{2}{3}$

$\alpha = 33.690\ldots^\circ = 33.7^\circ \quad (3\text{ s.f.}).$

This is above the positive horizontal for $k > 0$. If $k < 0$, the velocity points in the opposite direction.

a



b

Speed is the magnitude of velocity:

$(3\sqrt{13})^2 = (3k)^2 + (2k)^2$

$9 \times 13 = 9k^2 + 4k^2$

$117 = 13k^2 \quad \Rightarrow \quad k^2 = 9 \quad \Rightarrow \quad k = \pm 3.$

If projection is specified upwards and rightwards, only $k = 3$ applies.

Exercise 6C — Y2 Mechanics Pages 117–120 Solution 15

9. A stone is thrown at $30\,\text{m s}^{-1}$ from a window 20 m above horizontal ground. It hits the ground 3.5 s later. Find:

a. Its angle of projection.

b. The horizontal distance from the window to its landing point.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Let α be elevation above horizontal.

$u_x = 30\cos\alpha, \qquad u_y = 30\sin\alpha.$

Take upwards as positive:

$s = -20, \quad a = -9.8, \quad t = 3.5.$

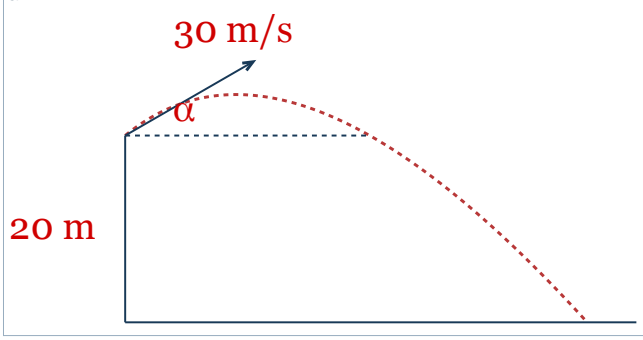
$s = ut + \frac{1}{2}at^2$

$-20 = 30\sin\alpha \times 3.5 - 4.9(3.5)^2$

$\sin\alpha = \dfrac{4.9(3.5)^2 - 20}{30 \times 3.5} = 0.381190\ldots$

$\alpha = 22.407\ldots^\circ \approx 22^\circ \quad (2\text{ s.f.}).$

a



b

Horizontally, $u = 30\cos 22.407\ldots^\circ$, $t = 3.5$:

$s = ut = 30\cos 22.407\ldots^\circ \times 3.5$

$= 97.072\ldots \approx 97\text{ m} \quad (2\text{ s.f.}).$

Exercise 6C — Y2 Mechanics Pages 117–120 Solution 16

10. A ball is thrown from O on horizontal ground at $U\,\text{m s}^{-1}$, at elevation θ, where $\tan\theta = 3/4$. It strikes a vertical wall 20 m from O, 3 m above the ground. Find:

a. U.

b. The time from projection to striking the wall.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

$\sin\theta = \dfrac{3}{5}, \quad \cos\theta = \dfrac{4}{5}, \quad u_x = \dfrac{4U}{5}, \quad u_y = \dfrac{3U}{5}.$

Horizontally:

$20 = \dfrac{4U}{5}t \quad \Rightarrow \quad t = \dfrac{25}{U}.$

Vertically upwards:

$s = 3, \quad a = -9.8, \quad s = ut + \frac{1}{2}at^2.$

$3 = \dfrac{3U}{5}t - 4.9t^2.$

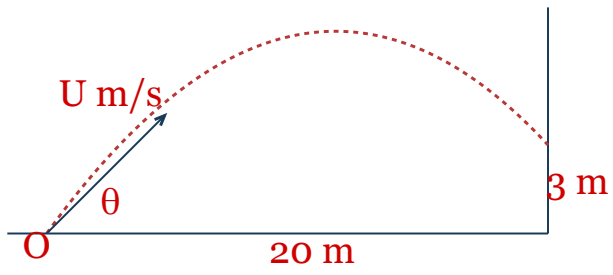
Substitute $t = 25/U$:

$3 = \dfrac{3U}{5}\dfrac{25}{U} - 4.9\dfrac{25^2}{U^2} = 15 - \dfrac{3062.5}{U^2}.$

$U^2 = \dfrac{3062.5}{12} = 255.208\ldots$

$U = 15.975\ldots \approx 16\,\text{m s}^{-1} \quad (2\text{ s.f.}).$

a



b

$t = \dfrac{25}{U} = \dfrac{25}{15.975\ldots} = 1.5649\ldots$

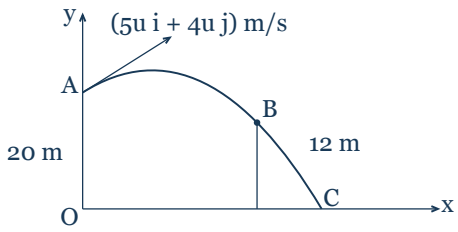
$t = 1.6\text{ s} \quad (2\text{ s.f.}).$

11. Particle P is projected from A, position $20\mathbf{j}$ m relative to fixed origin O, with velocity $(5u\mathbf{i} + 4u\mathbf{j})\ \mathrm{m\,s^{-1}}$. It moves freely under gravity through B, position $(k\mathbf{i} + 12\mathbf{j})$ m, before reaching C on the x-axis. Travel from A to B takes 4 s. Find:

a. u.

b. k.

c. The angle its velocity makes with the x-axis at C.



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

From A to B, upwards positive:

$u_y = 4u, \quad s = 12 - 20 = -8.$

$a = -9.8, \quad t = 4.$

$s = u_y t + \frac{1}{2}at^2$

$-8 = 4u \times 4 - 4.9 \times 4^2$

$u = \dfrac{4.9 \times 4^2 - 8}{16} = 4.4.$

b

Horizontally from A to B:

$u_x = 5u = 5(4.4) = 22, \quad t = 4.$

$s = u_x t \quad \Rightarrow \quad k = 22 \times 4 = 88.$

c

$v_x = 22$ throughout. At C:

$u_y = 4(4.4), \quad a = -9.8, \quad s = -20.$

$v_y^2 = u_y^2 + 2as$

$= (4 \times 4.4)^2 + 2(-9.8)(-20)$

$= 16(4.4)^2 + 392 = 701.76.$

The particle is descending:

$\tan\theta = \dfrac{|v_y|}{v_x} = \dfrac{\sqrt{701.76}}{22}.$

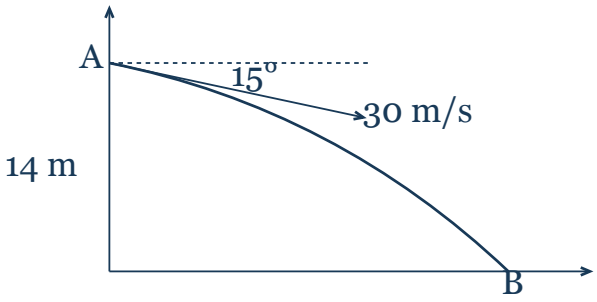
$\theta = 50.291\ldots^\circ \approx 50^\circ$

below the positive x-axis (2 s.f.).

12. A stone is thrown from A at $30\ \mathrm{m\,s^{-1}}$, 15° below horizontal. Point A is 14 m above horizontal ground. It strikes the ground at B. Find:

a. The time from A to B.

b. The distance AB.



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Take downwards as positive:

$u_x = 30\cos 15^\circ, \quad u_y = 30\sin 15^\circ.$

$s = 14, \quad a = 9.8.$

$s = u_y t + \frac{1}{2}at^2$

$14 = 30t\sin 15^\circ + 4.9t^2$

$4.9t^2 + 30t\sin 15^\circ - 14 = 0.$

The positive root is:

$t = \dfrac{-30\sin 15^\circ + \sqrt{900\sin^2 15^\circ + 4(14)(4.9)}}{9.8}$

$= 1.074\ldots \approx 1.1\ \mathrm{s} \quad (2\ \mathrm{s.f.}).$

b

Horizontally, $u_x = 30\cos 15^\circ$:

$s = u_x t = (30\cos 15^\circ)(1.074\ldots)$

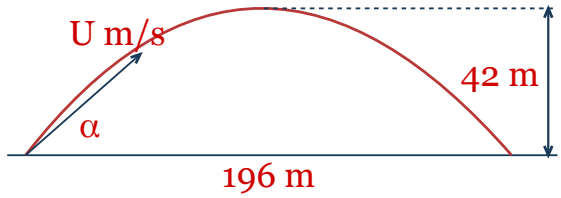
$= 31.136\ldots\ \mathrm{m}.$

$AB^2 = 14^2 + (31.136\ldots)^2 = 1165.456\ldots$

$AB = 34.138\ldots \approx 34\ \mathrm{m} \quad (2\ \mathrm{s.f.}).$

13. A particle is projected from level ground at $U\,\mathrm{m\,s^{-1}}$, at elevation α. Its maximum height is 42 m, and it lands 196 m from projection. Find α and U.

Working



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

Working

$u_x = U\cos\alpha$, $u_y = U\sin\alpha$. At maximum height, $v_y = 0$, $s = 42$:

$v_y^2 = u_y^2 + 2as$

$0 = U^2\sin^2\alpha - 2g(42)$

$U^2\sin^2\alpha = 84g. \qquad (1)$

Find two expressions for flight time. Horizontally:

$196 = U\cos\alpha\, t \quad \Rightarrow \quad t = \dfrac{196}{U\cos\alpha}.$

Vertically back at ground level:

$0 = Ut\sin\alpha - \frac{1}{2}gt^2 = t(U\sin\alpha - \frac{1}{2}gt).$

Ignore $t = 0$:

$t = \dfrac{2U\sin\alpha}{g}.$

Working

Equate the times:

$\dfrac{196}{U\cos\alpha} = \dfrac{2U\sin\alpha}{g}$

$U^2\sin\alpha\cos\alpha = 98g. \qquad (2)$

Divide (1) by (2):

$\tan\alpha = \dfrac{84g}{98g} = \dfrac{6}{7}$

$\alpha = 40.6^\circ \quad (3\text{ s.f.}).$

Substitute in (1), taking the positive speed:

$U\sin\alpha = \sqrt{84g}$

$U = \dfrac{\sqrt{84(9.8)}}{\sin\alpha} \approx 44\,\mathrm{m\,s^{-1}} \quad (2\text{ s.f.}).$

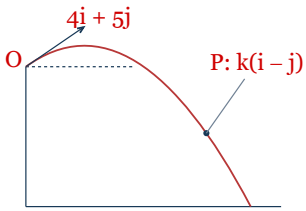
15. An object is projected from fixed origin O at $(4\mathbf{i} + 5\mathbf{j})\,\mathrm{m\,s^{-1}}$. It moves freely under gravity through P, position $k(\mathbf{i} - \mathbf{j})$ m, where $k > 0$.

a. Find k.

b.i. Find its speed at P.

b.ii. Find its direction of motion at P.

a



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Horizontally, $u_x = 4$, $s = k$:

$s = vt \quad \Rightarrow \quad k = 4t \quad \Rightarrow \quad t = k/4.$

Vertically at P, upwards positive:

$u_y = 5, \quad a = -9.8, \quad s = -k.$

$s = u_y t + \frac{1}{2}at^2$

$-k = \dfrac{5k}{4} - 4.9\dfrac{k^2}{16}.$

Divide by $k > 0$:

$\dfrac{9}{4} = \dfrac{4.9k}{16} \quad \Rightarrow \quad k = \dfrac{4\times 9}{4.9}$

$= 7.3469\ldots \approx 7.35 \quad (3\text{ s.f.}).$

b.i

At P, $v_x = 4$. Vertically:

$v_y^2 = u_y^2 + 2as = 5^2 + 2(-9.8)(-k) = 25 + 19.6k.$

$v^2 = v_x^2 + v_y^2 = 41 + 19.6k = 185.$

$v = \sqrt{185} = 13.601\ldots \approx 13.6\,\mathrm{m\,s^{-1}} \quad (3\text{ s.f.}).$

b.ii

The particle is descending. For the angle below horizontal:

$\cos\alpha = \dfrac{v_x}{v} = \dfrac{4}{\sqrt{185}}.$

Alternatively use $\tan\alpha = |v_y|/v_x$ or $\sin\alpha = |v_y|/v$.

$\alpha = 72.897\ldots^\circ \approx 72.9^\circ \quad (3\text{ s.f.})$

below the horizontal.

Exercise 6C — Y2 Mechanics Pages 117–120

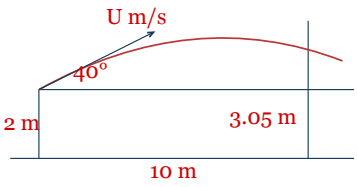
Solution 21

16. A basketball player stands on the floor 10 m from a basket at height 3.05 m. The ball is thrown from height 2 m, at 40° above horizontal. Model it as a particle moving in a vertical plane. It passes through the basket.

a. Find its initial speed.

b. State two factors ignored by modelling it as a particle.

a



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Let initial speed be U. Horizontally:

$u_x = U\cos 40°, \quad s = 10, \quad 10 = Ut\cos 40°$

$t = \dfrac{10}{U\cos 40°}.$

Vertically upwards, $u_y = U\sin 40°$, $a = -9.8$, $s = 3.05 - 2 = 1.05$:

$s = u_y t + \frac{1}{2}at^2$

$1.05 = \dfrac{10U\sin 40°}{U\cos 40°} - 4.9\left(\dfrac{10}{U\cos 40°}\right)^2$

$1.05 = 10\tan 40° - \dfrac{490}{(U\cos 40°)^2}$

$(U\cos 40°)^2 = \dfrac{490}{10\tan 40° - 1.05}$

$U^2 = \dfrac{490}{(10\tan 40° - 1.05)\cos^2 40°}$

$U = 10.665\ldots \approx 10.7\,\text{m s}^{-1} \quad (3\text{ s.f.}).$

b

The ball's size and its rotation/spin are ignored.

The projectile calculation also neglects air resistance. Weight is not ignored: it causes gravitational acceleration.

Extension — Maximum Range

Solution 22

Extension

A ball is projected from level ground with fixed speed u_0, at elevation θ. Prove its horizontal range is greatest at 45°.

Year 2 differentiation is required.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

Working

Take upwards as positive, $s = 0$, $u = u_0\sin\theta$, $a = -g$:

$0 = u_0 t\sin\theta - \frac{1}{2}gt^2 = t(u_0\sin\theta - \frac{1}{2}gt).$

Ignoring $t = 0$, flight time is $t = 2u_0\sin\theta/g$.

Horizontally:

$x = u_0\cos\theta\dfrac{2u_0\sin\theta}{g} = \dfrac{2u_0^2}{g}\sin\theta\cos\theta$

$x = \dfrac{u_0^2}{g}\sin 2\theta.$

Differentiate with θ in radians:

$\dfrac{dx}{d\theta} = \dfrac{2u_0^2}{g}\cos 2\theta = 0$

$2\theta = \dfrac{\pi}{2} \quad \Rightarrow \quad \theta = \dfrac{\pi}{4} = 45°.$

Here $d^2x/d\theta^2 = -4u_0^2/g < 0$, so this is a maximum for $0 < \theta < 90°$.

Projectile Formulae

Solution 23

You may be asked to prove general results using the same projectile methods.

A particle is projected from a horizontal plane at speed U, elevation α, and moves freely under gravity until it returns to the plane at B. Find expressions for:

a. The time of flight T.

b. The range R.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Take upwards as positive:

$s = 0, \quad u = U\sin\alpha, \quad a = -g.$

$s = ut + \frac{1}{2}at^2$

$0 = (U\sin\alpha)t - \frac{1}{2}gt^2 = t(U\sin\alpha - \frac{1}{2}gt).$

$t = 0$ at launch, $\quad T = \dfrac{2U\sin\alpha}{g}.$

b

Horizontally:

$$R = U\cos\alpha\frac{2U\sin\alpha}{g} = \frac{2U^2}{g}\sin\alpha\cos\alpha.$$

Use the double-angle formula for sine:

$$R = \frac{U^2\sin 2\alpha}{g}.$$

Projectile Formulae — Trajectory

Solution 24

A particle is projected at speed U, elevation α, and moves freely under gravity. After horizontal displacement x, its height above launch is y.

a. Show that

$$y = x\tan\alpha - \frac{gx^2}{2U^2}(1 + \tan^2\alpha).$$

A particle is projected from O on a horizontal plane at $28\,\text{m s}^{-1}$. It passes through B, $32\,\text{m}$ horizontally from O and $8\,\text{m}$ above the plane.

b. Find both possible elevations α, to the nearest degree.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Horizontally:

$x = U\cos\alpha\, t. \qquad (1)$

Vertically:

$y = U\sin\alpha\, t - \frac{1}{2}gt^2. \qquad (2)$

The target has no t. Make t the subject in (1), then substitute in (2):

$$t = \frac{x}{U\cos\alpha}$$

$$y = U\sin\alpha\frac{x}{U\cos\alpha} - \frac{g}{2}\left(\frac{x}{U\cos\alpha}\right)^2$$

$$= x\tan\alpha - \frac{gx^2}{2U^2}\sec^2\alpha$$

$$= x\tan\alpha - \frac{gx^2}{2U^2}(1 + \tan^2\alpha).$$

Do not be intimidated by symbols: resolve horizontally and vertically as usual.

Projectile Formulae — Two Possible Angles

Solution 25

A particle is projected at speed U, elevation α, and moves freely under gravity. After horizontal displacement x, its height above launch is y.

a. Show that

$$y = x\tan\alpha - \frac{gx^2}{2U^2}(1+\tan^2\alpha).$$

A particle is projected from O on a horizontal plane at $28\,\text{m s}^{-1}$. It passes through B, $32\,\text{m}$ horizontally from O and $8\,\text{m}$ above the plane.

b. Find both possible elevations α, to the nearest degree.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

b

$8 = 32\tan\alpha - 6.4(1+\tan^2\alpha).$

This is quadratic in $\tan\alpha$:

$6.4\tan^2\alpha - 32\tan\alpha + 14.4 = 0$

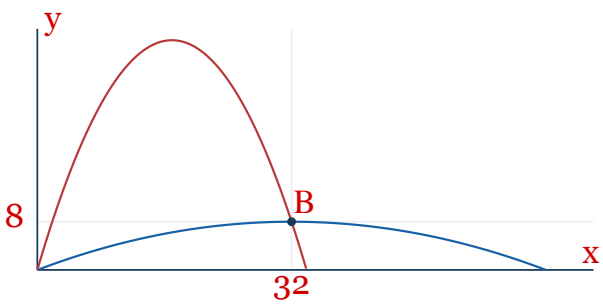
$4\tan^2\alpha - 20\tan\alpha + 9 = 0$

$(2\tan\alpha - 1)(2\tan\alpha - 9) = 0$

$\tan\alpha = \frac{1}{2}$ or $\frac{9}{2}$

$\alpha = 27^\circ$ or 77°.

b



b

For a fixed speed, there are generally two launch angles through a reachable point ahead. At the limiting reachable height, they coincide into one.

General Results

Solution 26

Exam note: you may be asked to derive these results. Use the resolving and suvat techniques from this chapter rather than memorising formulae for solving problems.

For initial speed U, elevation α, motion freely under gravity, returning to the same horizontal plane:

Time of flight $= \dfrac{2U\sin\alpha}{g}$.

Time to greatest height $= \dfrac{U\sin\alpha}{g}$.

Range $= \dfrac{U^2\sin 2\alpha}{g}$.

Trajectory: $y = x\tan\alpha - \dfrac{gx^2}{2U^2}(1+\tan^2\alpha)$.

Here y is height above launch and x is horizontal displacement.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

Exercise 6D — Y2 Mechanics Pages 123–125 Solution 27

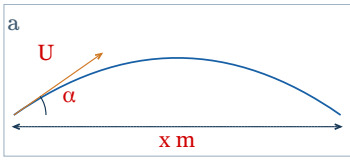
5. Use $g = 10\,\text{m s}^{-2}$ in this question. A particle is projected from O at speed U, elevation α, and moves freely under gravity. Its horizontal displacement is x, height above O is y.

a. Show that $y = x\tan\alpha - gx^2/(2U^2\cos^2\alpha)$.

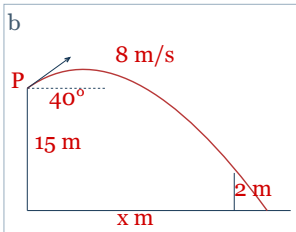
A stone is thrown from P at the end of a pier, 15 m above sea level, at $8\,\text{m s}^{-1}$ and elevation $40°$. Model it as a particle moving freely under gravity.

b. Find its horizontal distance from P when 2 m above sea level.

a



b



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Horizontally, $u_x = U\cos\alpha$:

$x = (U\cos\alpha)t \quad \Rightarrow \quad t = \dfrac{x}{U\cos\alpha}.$

Vertically upwards, $u_y = U\sin\alpha$, $a = -g$:

$y = u_y t + \frac{1}{2}at^2$

$y = U\sin\alpha\dfrac{x}{U\cos\alpha} - \dfrac{g}{2}\left(\dfrac{x}{U\cos\alpha}\right)^2$

$y = x\tan\alpha - \dfrac{gx^2}{2U^2\cos^2\alpha}.$

b

$U = 8$, $\alpha = 40°$, $y = 2 - 15 = -13$, $g = 10$:

$-13 = x\tan 40° - \dfrac{10x^2}{2(8^2)\cos^2 40°}.$

$0.13313\ldots x^2 - 0.83910\ldots x - 13 = 0.$

Let $a = 10/(128\cos^2 40°)$. The quadratic formula gives

$x = \dfrac{\tan 40° \pm \sqrt{\tan^2 40° + 52a}}{2a}.$

$x = 13.5234\ldots \quad \text{or} \quad -7.2206\ldots.$

Reject the negative root, which lies behind projection:

$x = 13.5234\ldots \approx 13.5\,\text{m} \quad (3\text{ s.f.}).$

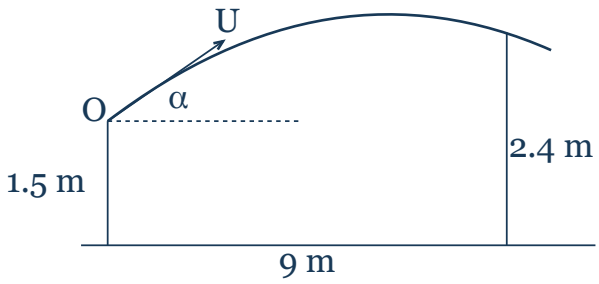
Exercise 6D — Y2 Mechanics Pages 123–125 Solution 28

7. A volleyball player hits a ball towards a net 9 m away, from height 1.5 m. It moves in a vertical plane perpendicular to the net and just passes over its top, 2.4 m above ground. Model it as a particle projected from O at $U\,\text{m s}^{-1}$, elevation α.

a. Write expressions for horizontal and vertical displacements after t seconds and show at the net that

$$0.9 = 9\tan\alpha - \frac{81g}{2U^2\cos^2\alpha}.$$

b. Given $\alpha = 30°$, find the speed as it passes over the net.



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

After t seconds:

$x = U\cos\alpha\, t, \qquad y = U\sin\alpha\, t - \frac{1}{2}gt^2.$

Horizontally:

$s = u_x t \quad \Rightarrow \quad 9 = U\cos\alpha\, t \quad \Rightarrow \quad t = \dfrac{9}{U\cos\alpha}.$

Vertically, $u_y = U\sin\alpha$, $a = -g$,
$s = 2.4 - 1.5 = 0.9$:

$0.9 = U\sin\alpha\dfrac{9}{U\cos\alpha} - \dfrac{g}{2}\left(\dfrac{9}{U\cos\alpha}\right)^2$

$0.9 = 9\tan\alpha - \dfrac{81g}{2U^2\cos^2\alpha}.$

b

$\tan 30° = \dfrac{1}{\sqrt{3}}, \quad \cos 30° = \dfrac{\sqrt{3}}{2}, \quad \sin 30° = \dfrac{1}{2}.$

$0.9 = \dfrac{9}{\sqrt{3}} - \dfrac{4(81)g}{2U^2(3)}$

$4.296\ldots = \dfrac{529.2}{U^2} \quad \Rightarrow \quad U = 11.098\ldots.$

At the net, $v_x = U\cos 30° = 9.6117\ldots$.

$v_y^2 = u_y^2 + 2as = (U/2)^2 - 2(9.8)(0.9)$

$= 13.15\ldots.$

$v^2 = v_x^2 + v_y^2 = 105.53\ldots$

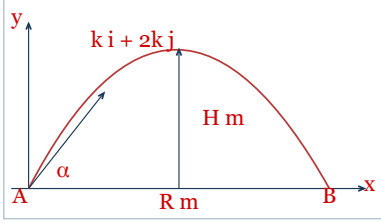
$v = 10.273\ldots \approx 10.3\,\text{m s}^{-1} \quad (3\text{ s.f.}).$

8. Unit vectors $\mathbf{i}$, $\mathbf{j}$ point horizontally and vertically upwards. An object is projected from A on horizontal ground at $(k\mathbf{i} + 2k\mathbf{j})\,\mathrm{m\,s^{-1}}$, $k > 0$. It moves freely under gravity until it lands at B and immediately stops. Take A as origin, with position $(x\mathbf{i} + y\mathbf{j})\,\mathrm{m}$.

a. Show that $y = 2x - gx^2/(2k^2)$.

b. Given range $AB = R$ and maximum height H, find in terms of k, g: i. R; ii. H.

a



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Horizontally, $u_x = k$:

$x = kt \quad \Rightarrow \quad t = x/k.$

Vertically, $u_y = 2k$, $a = -g$:

$y = u_y t + \frac{1}{2}at^2$

$y = \frac{2kx}{k} - \frac{g}{2}\left(\frac{x}{k}\right)^2$

$y = 2x - \frac{gx^2}{2k^2}.$

b.i

At landing, $x = R > 0$, $y = 0$:

$0 = 2R - \frac{gR^2}{2k^2} \quad \Rightarrow \quad \frac{gR^2}{2k^2} = 2R.$

$R^2 = \frac{4Rk^2}{g} \quad \Rightarrow \quad R = \frac{4k^2}{g}.$

b.ii

At maximum height, symmetry gives

$x = \frac{R}{2} = \frac{2k^2}{g}.$

Substitute into the trajectory:

$H = 2\frac{2k^2}{g} - \frac{g}{2k^2}\left(\frac{2k^2}{g}\right)^2$

$H = \frac{4k^2}{g} - \frac{2k^2}{g} = \frac{2k^2}{g}.$